Acidic extraction of histone proteins

Risk assessment

– Work with highly flammable liquids
– Work with concentrated acids
▷ Handle acids and flammable liquids in a certified chemical fume hood
▷ Wear gloves, safety glasses, lab coat
◻ Collect acetone as ORGANIC SOLVENT WASTE

Reviewed: Oct 26, 2023

This is a bench card. Full protocol available online.

›› *Acid extraction*

(1.) Start with a nuclei pellet or a nuclear fraction isolated from approximately 1.0×10^6–1.0×10^7 cells. Approximate the packed cell volume; this is 1 vol.

(2.) Gently vortex the pellet from the preceding nuclear isolation or extraction procedure.

(3.) *Critical:* In a dropwise manner, add 5 vol (10 vol if starting with fewer than 1×10^6 cells) of 0.2 M hydrochloric acid to the pellet at 4 °C. Gently pipette up and down to disperse the pellet.

Critical: Add slowly to avoid rapid precipitation. If working with small volumes, pipette gently rather than dropwise.

(4.) Incubate on ice for 5 min. 5 min

(5.) Centrifuge at 15 000 × *g* for 5 min at 4 °C. Transfer the supernatant to a clean tube.

(6.) To the supernatant, add 0.25 vol of 100% trichloroacetic acid. Vortex vigorously to precipitate histones.

Critical: Mix immediately and completely to avoid incomplete precipitation.

(7.) Precipitate on ice for 5 min. 5 min

(8.) Collect the histone pellet at 15 000 × *g* for 5 min at 4 °C. Discard supernatant.

(9.) Carefully add 100 µL cold (−20 °C) 99.9% acetone with HCl to the pellet. Do not disturb the pellet!

Quality assurance: Since acetone can degrade some plastics, perform washes quickly in compatible tubes.

(10.) Centrifuge at 15 000 × *g* for 2 min at 4 °C. Remove acetone using a glass pipette.

Quality assurance: Do not scrape the walls. Leave residual solvent rather than risk losing pellet material. A small amount of acetone will evaporate quickly.

(11.) Repeat acetone wash using 100 µL pure acetone (without HCl).

(12.) Air dry the pellet at room temperature.

Critical: Avoid overdrying. Desiccated pellets may be difficult to redissolve.

Storage of isolated histones

(1.) Store the histone pellet at room temperature for up to one week. For long-term storage, keep at −80 °C.

⊞ Optional: *Fractionation by reversed-phase HPLC*

(1.) Fractionate the histones by HPLC for long-term storage or mass spectrometry analysis.

(2.) Resuspend the dried histone pellet in 85 µL HPLC solvent A. 🕮

(3.) Inject the sample into a reverse-phase HPLC system equipped with a C18 or C3 column. 🕮

(4.) Separate histones using a linear gradient of HPLC solvent B starting at 20–30% B and running over 30–60 min. 🕮

> ***Critical:*** The linker histone H1 elutes first, ahead of the core histones, and H3 last: the order is H1, then H2B, then H2A and H4, then H3. Start collecting at the beginning of the gradient to keep H1, or only once the core histones begin if H1 is not wanted. Modified histones and histone variants elute between 15–45% B depending on sequence and modification, so a fraction is assigned by gel or mass spectrometry rather than by retention time alone. ⬅

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